

Homework 1

1.2.6 Let (X, μ) be a measure space and let E be a subset of X with $\mu(E) < \infty$. Assume that f is in $L^{p,\infty}(X, \mu)$ for some $0 < p < \infty$.

(a) Show for $0 < q < p$ we have

$$\int_E |f(x)|^q d\mu(x) \leq \frac{p}{p-q} \mu(E)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q.$$

Proof. We begin by rewriting $\int_E |f|^q d\mu$ with the distribution function. Note that proposition 1.2.3 requires the space over which we integrate to be σ -finite, however $\mu(E) < \infty$ and thus σ -finite.

$$\int_E |f|^q d\mu = \int_0^\infty q\lambda^{q-1} D_{f|_E}(\lambda) d\lambda = q \int_0^\infty \lambda^{q-1} \mu(\{x \in X : |f(x)| > \lambda\} \cap E) d\lambda.$$

Note that:

$$\begin{aligned} \|f\|_{L^{p,\infty}} &= \sup\{\lambda \mu(\{x : |f(x)| > \lambda\})^{\frac{1}{p}}\} \geq \lambda \mu(\{x : |f(x)| > \lambda\})^{\frac{1}{p}}. \\ \Rightarrow \|f\|_{L^{p,\infty}}^p &\geq \lambda^p \mu(\{x : |f(x)| > \lambda\}) \Leftrightarrow \mu(\{x : |f(x)| > \lambda\}) \leq \frac{\|f\|_{L^{p,\infty}}^p}{\lambda^p}. \end{aligned}$$

It is clear that $\mu(E)$ is an upper bound for $\mu(\{x : |f(x)| > \lambda\} \cap E)$. Additionally, $\mu(\{x : |f(x)| > \lambda\} \cap E) \leq \mu(\{x : |f(x)| > \lambda\})$. So,

$$q \int_0^\infty \lambda^{q-1} \mu(\{x : |f(x)| > \lambda\} \cap E) d\lambda \leq q \int_0^\infty \lambda^{q-1} \min(\mu(E), \frac{\|f\|_{L^{p,\infty}}^p}{\lambda^p}) d\lambda.$$

We evaluate this integral by splitting where $\mu(E) = \|f\|_{L^{p,\infty}}^p / \lambda^p$. With some algebra, this ends up being at $\lambda = \|f\|_{L^{p,\infty}} \cdot \mu(E)^{-\frac{1}{p}}$. We'll call it λ_0 .

$$\begin{aligned} \int_0^\infty q\lambda^{q-1} \min(\mu(E), \frac{\|f\|_{L^{p,\infty}}^p}{\lambda^p}) d\lambda &= \int_0^{\lambda_0} q\lambda^{q-1} \mu(E) d\lambda + \int_{\lambda_0}^\infty q\lambda^{q-1} \frac{\|f\|_{L^{p,\infty}}^p}{\lambda^p} d\lambda. \\ &= \int_0^{\lambda_0} q\lambda^{q-1} \mu(E) d\lambda + \int_{\lambda_0}^\infty q\lambda^{q-1-p} \|f\|_{L^{p,\infty}}^p d\lambda. \\ &= \lambda_0^q \mu(E) + \frac{q}{p-q} \lambda_0^{q-p} \|f\|_{L^{p,\infty}}^p. \end{aligned}$$

We now plug in λ_0 :

$$\begin{aligned} &[\|f\|_{L^{p,\infty}} \cdot \mu(E)^{-\frac{1}{p}}]^q \mu(E) + \frac{q}{p-q} [\|f\|_{L^{p,\infty}} \cdot \mu(E)^{-\frac{1}{p}}]^{q-p} \|f\|_{L^{p,\infty}}^p. \\ &= \|f\|_{L^{p,\infty}}^q \cdot \mu(E)^{-\frac{q}{p}+1} + \frac{q}{p-q} \|f\|_{L^{p,\infty}}^q \cdot \mu(E)^{\frac{p-q}{p}} = (1 + \frac{q}{p-q}) \mu(E)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q. \end{aligned}$$

Note that: $1 + \frac{q}{p-q} = \frac{p-q+q}{p-q} = \frac{p}{p-q}$, which gives:

$$\int_E |f|^q d\mu \leq \frac{p}{p-q} \mu(E)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q, \text{ as desired.}$$

□

(b) Show that $L^p(X, \mu) \subseteq L^{p,\infty}(X, \mu) \subseteq L^q(X, \mu)$, given that $\mu(X) < \infty$ and $0 < q < p < \infty$.

Proof. We wish to show that $L^{p,\infty}(X, \mu) \subseteq L^q(X, \mu)$. We know from part (a) that:

$$\int_E |f|^q d\mu \leq \frac{p}{p-q} \mu(E)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q$$

given $0 < q < p < \infty$ and $\mu(E) < \infty$. Because $\mu(X) < \infty$, we can let $E = X$.

$$\begin{aligned} \int_X |f|^q d\mu &\leq \frac{p}{p-q} \mu(X)^{1-\frac{q}{p}} \|f\|_{L^{p,\infty}}^q \\ \Rightarrow \left(\int_X |f|^q d\mu \right)^{\frac{1}{q}} &= \|f\|_{L^q} \leq \left(\frac{p}{p-q} \right)^{\frac{1}{q}} \mu(X)^{\frac{1}{q}-\frac{1}{p}} \|f\|_{L^{p,\infty}}. \end{aligned}$$

Note that p, q are fixed, and so, $\left(\frac{p}{p-q} \right)^{\frac{1}{q}} \mu(X)^{\frac{1}{q}-\frac{1}{p}}$ is a positive, finite constant, call it α . so:

$$\|f\|_{L^q} \leq \alpha \|f\|_{L^{p,\infty}}.$$

Therefore if $\|f\|_{L^{p,\infty}} < \infty$, then $\|f\|_{L^q} < \infty$. Thus, $L^{p,\infty}(X, \mu) \subseteq L^q(X, \mu)$.

The fact that $L^p(X, \mu) \subseteq L^{p,\infty}(X, \mu)$ is proved as proposition 1.2.6 in the text.

□

(c) Show that $L^{p,\infty}(\mathbb{R}^n, |\cdot|) \subseteq L^1_{\text{loc}}(\mathbb{R}^n, |\cdot|)$ whenever $p > 1$.

Proof. By Theorem 1.2.10 in the text, we have:

$$\|f\|_{L^{p,\infty}} \leq \sup_{0 < \mu(E) < \infty} \{ \mu(E)^{-1+1/p} \int_E |f| d\mu \} \leq \frac{p}{p-1} \|f\|_{L^{p,\infty}}.$$

If we now let E be a compact subset of $\mathbb{R}^n$, hence $\mu(E) < \infty$, we have that:

$$\mu(E)^{-1+1/p} \int_E |f| d\mu \leq \frac{p}{p-1} \|f\|_{L^{p,\infty}}.$$

If we let $\alpha = \frac{p}{p-1} \mu(E)^{1-1/p}$:

$$\int_E |f| d\mu \leq \alpha \|f\|_{L^{p,\infty}}.$$

So,

$$\|f\|_{L^{p,\infty}} < \infty \Rightarrow \int_E |f| d\mu \leq \alpha \|f\|_{L^{p,\infty}} < \infty.$$

Thus, f in $L^{p,\infty}(\mathbb{R}^n, |\cdot|)$ means that f is locally integrable on $\mathbb{R}^n$. So,

$$L^{p,\infty}(\mathbb{R}^n, |\cdot|) \subseteq L^1_{\text{loc}}(\mathbb{R}^n, |\cdot|).$$

□

Additional Problems

1.2.1 Suppose $f_k \geq 0$ are measurable functions on (X, μ) . Prove that if $f_k \rightarrow f$ from below as $k \rightarrow \infty$ a.e., then $D_{f_k} \rightarrow D_f$ from below as $k \rightarrow \infty$.

Proof. We first show that $D_{f_k} \leq D_f$ for all k . Fix λ . Consider $D_f(\lambda) = \mu(\{x : |f(x)| > \lambda\})$. Per assumption, $f_k(x) \leq f(x)$ for all x . Thus, $f_k(x) > \lambda \Rightarrow f(x) > \lambda$. Therefore we have that:

$$\{x : |f_k(x)| > \lambda\} \subseteq \{x : |f(x)| > \lambda\}.$$

and so:

$$\mu(\{x : |f_k(x)| > \lambda\}) \leq \mu(\{x : |f(x)| > \lambda\}).$$

Which is equivalent to $D_{f_k} \leq D_f$. Now we must show convergence of D_{f_k} to D_f .

Notice that because f_k are increasing and approach f from below, there exists a subsequence, f_{k_n} that is monotone increasing and approaches f from below. Thus,

$$\{x : |f_{k_n}(x)| > \lambda\} \subseteq \{x : |f_{k_{n+1}}(x)| > \lambda\}.$$

Therefore we can write $\{x : |f_{k_n}(x)| > \lambda\} = \bigcup_{t \leq n} \{x : |f_{k_t}(x)| > \lambda\}$. Thus,

$$\mu(\{x : |f_{k_n}(x)| > \lambda\}) = \mu\left(\bigcup_{t \leq n} \{x : |f_{k_t}(x)| > \lambda\}\right).$$

We apply continuity of measure from below to get:

$$\lim_{n \rightarrow \infty} \mu(\{x : |f_{k_n}(x)| > \lambda\}) = \mu\left(\bigcup_{n \in \mathbb{N}} \{x : |f_{k_n}(x)| > \lambda\}\right) = \mu(\{x : |f(x)| > \lambda\}).$$

Which is equivalent to:

$$\lim_{n \rightarrow \infty} D_{f_{k_n}} = D_f$$

□